MIL-C-43222B
27 August 1981
SUPERSEDING
MIL-C-43222A
21 December 1970

MILITARY SPECIFICATION

CHAIR, FOLDING, WOOD, WITH KNEELER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type wood folding chair, with kneeler.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC-A-680	-	Artificial Leather (Cloth Coated), Vinyl Resin, Expandable Layer, (Upholstery)
MMM-A-100	-	Adhesive, Animal Glue
MMM-A-125	-	Adhesive, Casein Type, Water and Mold Resistant
MMM-A-193	-	Adhesive, Vinyl Acetate Resin Emulsion
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-R-5001 - Rubber, Cellular Sheet, Molded and Hand Built Shapes: Latex Foam

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment
MIL-STD-1363 - Measure of Wood Moisture Content

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE

Commercial Standard CS 35-61 - Hardwood Plywood

Product Standard PS 1-66 - Softwood Plywood/Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The chair shall be the manufacturer's standard product, except for any changes necessary to comply with this specification.

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* 3.2 Materials. Materials (see 6.4) not definitely specified shall be of the quality normally used by the manufacturer for wood folding chairs with kneelers, provided the completed item complies with all provisions of this specification.

3.2.1 Lumber. Lumber shall be well seasoned, straight grained, beech, birch, maple, magnolia, or rock elm, uniformly dried without brashness, casehardening, or honeycombing, and shall be free from knots, decays, mineral streak, holes, wane, and shake. At time of assembly, of the wood components of the chair with kneeler moisture content shall be not less than 5 nor more than 9 percent, when tested as specified in 4.2.

3.2.2 Plywood. The plywood shall be of the thickness specified herein and shall be either softwood plywood or hardwood plywood.

3.2.2.1 Softwood plywood. Softwood plywood shall conform to group 1, interior types, grade C-D, plugged or better, sanded two sides of PS 1-66.

3.2.2.2 Hardwood plywood. All hardwood plywood panels shall be grade 1, high density face and back veneer conforming to type II of CS 35-61.

3.2.3 Adhesives. The adhesives shall conform to type I, grade J2 or better, or type II, grade A1 or B1 of MMM-A-100; type I of MMM-A-125; or to MMM-A-193.

3.2.4 Rubber, latex. Latex rubber shall conform to type II, class firm, grade B of MIL-R-5001.

* 3.2.5 Artificial leather. The artificial leather shall conform to class 2, treatment (a) 1, (b) or (c), of CCC-C-680, color to match No. 33578 of FED-STD-595.

3.3 Design. The structural design shall be such that any sustained or impact weight shall be uniformly distributed on all supporting parts with no excessive strain under normal usage. The chair shall be designed to provide for folding and shall form a side-type chair with all legs in a vertical position when set-up. Folding shall be accomplished by raising either the front or the rear of the seat section, which shall cause the upholstery top of the seat or the bottom of the seat to fold against the back and cause the front legs to fold parallel with the rear legs. The kneeler shall be constructed to fold between the rear legs and secured to prevent accidentally falling. At the contractor's option, the kneeler may fold flat against the rear legs. The folding mechanism shall be designed to guard against injury and snagging of the user's clothing. When in use, the chair shall not collapse due to a shift in weight of the user. The design shall be such that one person can unfold and fold the chair. The chair shall fold to permit stacking directly over one another in a free standing pile, twelve chairs high, when tested as specified in 4.2.4.

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3.4 Construction. The chair shall conform to the requirements specified herein and in table I. The chair shall consist of the back section, seat section, front legs sections, kneeler, and the folding mechanism. The parts of the folding mechanism shall be jointed together. Chair shall be made of like species of wood throughout, except for the seat pad and kneeler pad. All legs shall be equipped with 3/4 ± 1/64 inch diameter hard rubber, single prong glides.

TABLE I. Physical requirements (in inches)

Height overall	32-1/2 ± 1
Width of back	14-15/16 ± 1/2
Front	Not less than 15-1/4
Depth of seat frame	Not less than 14-1/2
Height of seat from floor	17-1/2 ± 1/2
Width of back	Not less than 12-3/4
Depth of kneeler frame	12-1/4 ± 1/2
Height to top of kneeler pad	Not less than 6
Width of kneeler frame	Not less than 12

3.4.1 Seat frame (rails). Seat frames shall be not less than 3/4 inch thick, jointed at corners by tongue and groove joints, full width of material, or by miter and spline, and glued. Alternatively, the frames may be steam bent and attached to a notched headerblock by screws and glue.

3.4.2 Back posts. The back posts shall be not less than 3/4-inch thick or less than 1-1/8 inches wide at narrowest point. Posts may be steam bent or band sawed to shape. Spindles, slats, or back rests shall be not less than 11/16 inch wide and shall be steam bent or sawed. Horizontal spindles, when utilized, shall be mortised into back post vertical spindles. Slats or back rests, when utilized, shall be mortised into a top and bottom rail.

3.4.3 Rungs or stretchers. The rungs or stretchers shall be not less than 11/16 inch square, finished, and shall be joined to the legs by mortise and tenon joints, and glued. There shall be at least one rung or stretcher placed in the front leg assembly. If the kneeler folds flat against the rear legs, there shall be at least one rung or stretcher placed between the rear back posts.

3.4.4 Tie rod or hinge for kneeler. If the kneeler folds between the rear legs, the tie rod or hinge shall be 5/16 ± 1/64 inch diameter with spacers positioned and riveted so as to function as a brace for the back legs and hinge for the kneeler. If the kneeler folds flat against the rear legs, the folding mechanism or offset hinge shall be hot rolled steel at least 1/2 inch wide and 0.075 inch thick with riveted joints and attached to the rear legs of the chair which shall lock against the chair when folded.

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3.4.5 Front legs. The front legs shall be not less than 13/16 inch thick nor less than 7/8 inch wide at the bottom and shall be straight, shaped, or tapered.

3.4.6 Joints. All joints shall be glued with any of the adhesives specified in 3.2.3.

* 3.4.7 Seat pad. The seat pad shall be constructed with not less than 5/16 inch thickness plywood bottom padded with polyurethane (see 6.3), forming a 1 inch high crown and covered with artificial leather specified in 3.2.5. The artificial leather of the seat pad shall be attached to permit removal and replacement without the use of special tools.

3.4.8 Kneeler pad. The kneeler pad shall be latex foam rubber specified in 3.2.4 not less than 1 inch thick on a plywood panel not less than 3/8 inch thick. The foam rubber shall be covered with artificial leather. The artificial leather shall be attached in the same manner as the seat pad.

3.5 Finish.

3.5.1 Preparation for finish. Prior to finishing, all exposed wood surfaces shall have been thoroughly sanded smooth. Glue size may be used before this sanding in order to stiffen wood fibers. Surface shall be thoroughly cleaned immediately before application of stain.

3.5.2 Application of finish. All wood surfaces of the chair shall be stained, sealed, and satin gloss lacquered or varnished to a light oak or sand finish. Chairs shall be finished with not less than two coats of lacquer or varnish and shall be sanded between coats in accordance with good commercial practice. All metal parts shall be enameled in a color to harmonize with chair finish.

3.6 Marking for identification. The contractor shall permanently and legibly affix his name, or trademark of such character as to be identified with said contractor, on the underside of each chair.

3.7 Workmanship. The chair with kneeler shall be free of blisters, peeled or chipped areas, areas of no film, foreign matter embedded in finish, color separation, and poor touch-ups. Components shall be free of fractures, splits, dents, slivers, sharp burrs, and torn grain. Artificial leather covers shall be free of tears and rips and shall not be soiled.

3.7.1 Fabrication. The chair shall be fabricated in proper alignment, and components shall be finished and assembled. All sharp burrs, slivers, or splinters shall be removed.

3.7.2 Threaded fasteners. All bolts, nuts, screws, and studs shall be drawn up tight.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

* 4.2.2.1 In-process examination. Examination shall be made to determine that the following are as specified.

a. That the lumber is as specified in 3.2.1.
b. That the polyurethane foam is as specified in 3.4.7.
c. That gluing is accomplished as specified in 3.4.6.
d. That the plywood thickness is as specified in 3.4.7 and 3.4.8, as applicable.
e. That the latex foam rubber pad thickness is as specified in 3.4.8.
f. That sanding of exposed wood surfaces is accomplished as specified in 3.5.1 and 3.5.2.

Whenever nonconformance is noted, correction shall be made to the items affected and the process.

* 4.2.2.2 In-process testing. The moisture content of the solid lumber components, at time of assembly, shall be determined in accordance with MIL-STD-1363 to determine compliance with 3.2.1. The sample unit shall be one piece of each different component. The lot shall consist of the number of the same components prepared under the same conditions, stored as a group and offered for inspection at one time. The test shall be made immediately prior to assembly and finish. There shall be one determination made per unit. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 4.0 defects per hundred units. When testing is performed by the electric moisture meter method, three determinations shall be made on each sample unit. The average of the three determinations shall be the moisture content of sample unit.

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* 4.2.3 End item examination. The end item shall be examined in accordance with 4.2.3.1 and 4.2.3.2. The lot size shall be expressed in units of chairs. The sample unit shall be one chair.

* 4.2.3.1 Visual examination. The end item shall be examined for the defects listed in table II. The inspection level shall be II and the AQL shall be 2.5 major and 6.5 total (major and minor combined) defects per hundred units.

TABLE II. Visual defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as specified	X	
	Type or color of finish not as specified		X
	Blistered, peeled, chipped, area of no film		X
	Poor touch-up, color separation, stained, overspray, or foreign matter imbedded in finish		X
Design	Characteristics not in accordance with the specified requirements	X	
Construction and workmanship	Component missing or not specified type	X	
	Set or kneeler pad not covered with artificial leather	X	
	Chair not made of like species of wood throughout (except where otherwise specified)	X	
	Artificial leather not replaceable without the use of special tools	X	
	Component fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, or malformed	X	
	Part misplaced or loose	X	
	Sharp burr or sliver	X	
	Covers soiled, ripped, or torn		X
	Functioning component that is inoperative	X	
	Functioning component that requires abnormal force to operate		X
	Components not connected or joined as specified	X	
	Any threaded fasteners missing, not type specified, broken, or stripped	X	

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TABLE II. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly	Any component not in proper alignment; chair and kneeler in open position rocks more than 1/16 inch when placed on a smooth flat surface	X	
Marking	Missing, incomplete, illegible	X	

* 4.2.3.2 Dimensional examination. The end item shall be examined for compliance with the specified dimensions. Any deviation from the specified dimensions shall be considered a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

* 4.2.4 End item testing. The end item shall be tested as specified in 4.4.1 for conformance to the stacking requirement in 3.3. Regardless of lot size, the sample size shall be 12 chairs selected at random from the lot. Failure of this test shall be cause for rejection of the lot.

* 4.2.5 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Packaging defects

Examine	Defect
Markings	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Fiberboard pads missing. Contents not immobilized within shipping container.
Contents	Number per container is more or less than required.

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4.2.6 Palletization examination. An examination shall be made to determine that palletization complies with the section 5 requirements. Defects shall be scored in accordance with table IV. The sample unit shall be one palletized unit load ready for shipment. The lot size shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

TABLE IV. Palletization defects

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

* 4.3 Methods of inspection.

4.3.1 Stacking test. The twelve chairs shall be folded and stacked in a free standing pile one directly over another. Collapse of the pile shall constitute failure of this test (see 3.3).

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Four folded chairs shall be nested in the most compact manner. Each chair shall be slipsheeted with minimum 40-pound basis weight (24 by 36-500) draft paper or supported with fiberboard pads and cushioned as required to prevent movement or damage while in transit.

* 5.1.2 Commercial. Chairs shall be preserved in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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* 5.2.1 Level A packing. Four chairs, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Each shipping container shall be provided with two fiberboard pads, minimum bursting strength of 275 pounds per square inch (p.s.i.), one positioned at each end of the container. Additional fiberboard inserts shall be used, as required, to immobilize the contents. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

* 5.2.2 Level B packing. Four chairs, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be provided with two fiberboard pads, minimum bursting strength of 275 p.s.i., one positioned at each end of the container. Additional fiberboard inserts shall be used, as required, to immobilize the contents. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be a grade V3, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix thereto.

* 5.2.3 Commercial packing. Chairs, preserved as specified in 5.1, shall be packed in accordance wth MIL-STD-1188.

* 5.3 Palletization. When specified (see 6.2), chairs packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129, or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The wooden folding chair kneeler is intended for providing additional seating in Military chapels and seating in Blessed Sacrament rooms.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(d) When palletization is required (see 5.3).

* 6.3 Commercial product. The polyurethane foam conforming to Firestone Foamex 280 Polyurethane or equal, has been found to meet the requirements of 3.4.7.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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Army - GL
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Preparing activity:

Army - GL

Project No. 9925-0205

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-43222B Chair, Folding, Wood, With Kneeler

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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